SCOB032: COMPUTER NETWORKS

MASTER EXAM & TESTS PREP GUIDE: ASSESSMENT SCOPE, TEXTBOOK MAPPINGS & QUESTION BANK (2016–2026)

Comprehensive Course Roadmap & Exam Strategy: This document merges the complete assessment scope for Term Test 1, Term Test 2, Special Test 2, and the Final Examination under Dr. R.L. Moila. It aligns all lecture slide decks with prescribed textbook chapters (Tanenbaum 5e, Forouzan 5e, Stallings 9e/10e, Kurose & Ross 7e/8e) and contains the complete past question bank extracted directly from official 2016–2026 test and examination papers.

SECTION 1: OFFICIAL ASSESSMENT SCOPE (TESTS & FINAL EXAM)

1.1 Term Test 1 Scope (50 Marks | 1.5–2 Hours)

- **System Characteristics & Criteria:** Four system upgrade characteristics (Fault Tolerance, Scalability, QoS, Security) and three effective network criteria (Performance, Reliability, Security).
- **Topologies & Cable Vulnerabilities:** Bus, Star, Ring, Mesh topologies; bus topology central cable breaks and signal termination risks; selecting Star topologies with Layer 2 switches for independent departmental traffic (HR, IT, Marketing).
- **Transmission Media & Hazards:** Copper (Twisted Pair/Coaxial) vs Fiber Optics; attenuation, EMI, lightning/thunderstorm susceptibility; microwave line-of-sight limitations over mountainous terrain.
- **OSI Layer 1–4 Troubleshooting:** Layer 1 attenuation/weak signal; Layer 2 intra-subnet MAC delivery vs inter-subnet limitations; Layer 3 default gateway/inter-network routing failures; Layer 4 port mapping errors.
- **Switching & Hardware:** Circuit vs Message vs Packet Switching; Hub flooding vs Switch MAC learning; error detection (Parity) vs error correction (Hamming Code); Extranet B2B collaboration benefits, challenges, and implementation.

1.2 Term Test 2 & Special/Sick Test 2 Scope (50 Marks | 1.5–2 Hours)

- **Manual VLSM Subnetting (No Calculators):** Subnetting parent blocks (10.10.0.0/16, 172.20.0.0/23, 172.16.0.0/16, 192.168.10.0/24) using strictly descending host allocations.
- **Subnet Metrics:** CIDR prefix length, Subnet Mask (dotted-decimal), Subnet ID, First Usable IP, Last Usable IP, Broadcast Address, unallocated IP space.
- **Junior Technician Traps:** Evaluating flawed proposals (assigning Broadcast IDs, misallocating host ranges between subnets, FLSM /27 IP waste across unequal subnets).
- **Transport Layer & Connection Management:** TCP vs UDP state/reliability trade-offs; TCP 3-way handshake Sequence/ACK number math with non-zero ISNs (ISN_A, ISN_B); standard port mappings (HTTP 80, HTTPS 443, SSH 22, SMTP 25, DNS 53, FTP 21).
- **Wireless & Mobile Security:** Cellular base station handoff/BSS transitions; CSMA/CA vs CSMA/CD; Hidden Terminal Problem and RTS/CTS frame exchanges; Asymmetric Encryption (encrypt with Recipient Public Key, decrypt with Recipient Private Key); DoS/DDoS flooding, CIA Triad Availability, and proactive firewall/rate-limiting controls.

1.3 Final Main & Supplementary Examination Scope (60 Marks | 2 Hours)

- **Domain 1 (Physical Layer & Fundamentals):** Topologies, guided/unguided media, attenuation, EMI, 6 throughput factors, QAM, WDM, Bit Error Rate (BER) analysis.
- **Domain 2 (Data Link & MAC Protocols):** CSMA/CD, CSMA/CA, Pure/Slotted ALOHA, TDMA, Polling, Hidden Terminal / RTS-CTS, Hubs vs Switches vs Bridges, error detection/correction.
- **Domain 3 (Network Layer & Routing):** Binary IPv4 class derivations, FLSM vs VLSM, CIDR notation, Direct vs Indirect Mobile IP Routing (Home/Foreign Agents, Care-of Address, Triangle Routing Latency), MANET / DSR routing.
- **Domain 4 (Transport Layer):** TCP vs UDP, 3-way handshake Sequence/ACK math, packet reordering, congestion vs flow control, service port mappings.
- **Domain 5 (Application & Security):** DNS resolution, DHCP DORA process, Asymmetric Public Key Cryptography, Zero-Day vulnerabilities, DoS/DDoS mitigation, Passive vs Active attacks, CIA Triad.

SECTION 2: CHAPTER & TEXTBOOK PAGE MAPPINGS

Lecture Slide Deck	Core Topic Covered	Tanenbaum (5e)	Forouzan (5e)	Stallings (9e/10e)	Kurose & Ross
SECTION 1 COMPUTER NETWORKS.pptx	Architecture, Topologies, OSI & TCP/IP Reference Models	Ch. 1 (pp. 1–90)	Ch. 1–2 (pp. 1–56)	Ch. 1–2 (pp. 25–68)	Ch. 1 (pp. 1–88)
Physical layer.pptx & COMMUNICATION CHANNEL.pptx	Media, Throughput, Attenuation, EMI, BER, QAM	Ch. 2 (pp. 91–174)	Ch. 3–6 (pp. 57–190)	Ch. 3–5 (pp. 69–160)	Ch. 1 (pp. 50–65)
Datalink layer.pptx & MEDIUM ACCESS PROTOCOLS...	Framing, Parity/Hamming, CSMA/CD, CSMA/CA, ALOHA	Ch. 3–4 (pp. 175–342)	Ch. 9–14 (pp. 265–420)	Ch. 6–7, 15 (pp. 161–220)	Ch. 6 (pp. 468–530)
IP_Addressing_and Subnetting.pptx & SUBNETTING.pptx	IPv4 Classes, FLSM, VLSM, Dotted Decimal & CIDR	Ch. 5 (pp. 343–480)	Ch. 18–19 (pp. 547–610)	Ch. 18 (pp. 580–630)	Ch. 4 (pp. 330–380)
Network_Layer.pptx .pdf	Routing Algorithms, Direct vs Indirect Mobile IP	Ch. 5 (pp. 380–450)	Ch. 20–22 (pp. 611–708)	Ch. 19 (pp. 631–670)	Ch. 4–5 (pp. 381–467)
Chapter 5 Transport layer.pptx	TCP/UDP, 3-Way Handshake, Sequence/ACK Math, Ports	Ch. 6 (pp. 481–586)	Ch. 23–24 (pp. 709–780)	Ch. 20 (pp. 671–720)	Ch. 3 (pp. 185–300)
Chapter6-ApplicationLayer.pptx	DNS, DHCP DORA, HTTP, SMTP, FTP	Ch. 7 (pp. 587–712)	Ch. 25–27 (pp. 781–860)	Ch. 24 (pp. 810–860)	Ch. 2 (pp. 89–184)
MOBILE NETWORKS.pptx & MOBILE AND WIRELESS...	Hidden Terminal, RTS/CTS, MANET/DSR, Crypto, DDoS	Ch. 4, 8 (pp. 290–310)	Ch. 15, 31–32 (pp. 421–460)	Ch. 14, 21–22 (pp. 420–469)	Ch. 7–8 (pp. 531–670)

SECTION 3: EXHAUSTIVE PAST QUESTION BANK (2016–2026)

PART A: TERM TEST 1 PAST QUESTIONS (2024 & 2026)

1. SCOB032 Test 1 (2026) [50 Marks]

Q1 [10 Marks — OSI Troubleshooting Scenario]: A university computer laboratory experiences three problems: Computer A can communicate locally but cannot reach outside networks; Computer B receives signals that are too weak for reliable communication; Computer C receives data correctly, but it is delivered to the wrong application.
- **a)** Identify the most appropriate OSI layer to investigate for each computer (A–C).
- **b)** Identify one function of that layer relating directly to each problem.
- **c)** Explain why troubleshooting only the Data Link layer would not solve all three problems.

Q2 [10 Marks — Transmission Media Selection]: A company connects two offices located 25 km apart across tall buildings and mountainous terrain, evaluating twisted pair, fiber optic, and microwave transmission.
- **a)** Identify the major technical requirement that makes microwave transmission unsuitable for this scenario.
- **b)** Recommend the most appropriate physical medium and justify using three characteristics.
- **c)** If remote mountainous areas prevent cable installation, state how a microwave connection could be established.

Q3 [10 Marks — Switching Techniques]: A hospital network handles Service 1 (Real-time voice communication) and Service 2 (Overnight patient backup file transfers).
- **a)** Recommend the most appropriate switching technique for Service 1 with two reasons.
- **b)** Recommend the most appropriate switching technique for Service 2 with two reasons.
- **c) The Technician Trap:** A technician argues: *'Message switching should be used for Service 1 because intermediate nodes store and forward*

the complete message.' Critically evaluate this recommendation.

Q4 [10 Marks — Hardware & Network Segmentation]: A company has a 40-computer LAN connected through hubs, experiencing severe performance degradation.
• **a)** Explain why installing additional repeaters will not solve the performance problem.
• **b)** Evaluate three proposals (Technician A: Install repeaters; Technician B: Replace hubs with Layer 2 switches; Technician C: Install a router between every computer) and recommend the best choice.
• **c)** If the company creates a second IP subnet for another department, state whether a Layer 2 switch alone is sufficient to route traffic between them.

Q5 [10 Marks — Shared Media & Wireless Collision Logic]: Network X is an older half-duplex Ethernet network; Network Y is a Wi-Fi network with laptops A and B separated by a thick wall.
• **a)** Explain how CSMA/CD operates on Network X.
• **b)** Explain the Hidden Terminal Problem occurring on Network Y.
• **c)** Describe how CSMA/CA with RTS/CTS mitigates hidden terminal collisions on Network Y.
• **d) The Student Trap:** A student proposes replacing CSMA/CA on Network Y with CSMA/CD, arguing devices can simply detect collisions after they occur. Explain why this is inappropriate for wireless media.

2. SCOB032 Term Test 1 (2024) [50 Marks]

Q1.1 [8 Marks]: State and explain four fundamental characteristics that should be considered when upgrading an online learning platform.

Q1.2 [6 Marks]: Discuss the three criteria necessary for developing an effective and efficient network.

Q1.3 [5 Marks]: Identify and explain the most suitable network topology for a small office environment consisting of IT, HR, and Marketing departments.

Q1.4 [12 Marks — Extranet Integration]:
• **1.4.1** Explain, with an example, how ABC Bank's extranet improves business partner coordination.
• **1.4.2** State two benefits derived from extranet implementation.
• **1.4.3** Describe two real-world examples of companies utilizing extranets.
• **1.4.4** Deduce two challenges in designing and implementing an extranet.

Q1.5 [6 Marks]: A line manager can communicate with his local printer but cannot reach a manager's computer on another subnet. Explain how to begin troubleshooting using OSI layers.

Q1.6 [7 Marks — Bus Topology Failure]:
• **1.6.1** What happens if the central cable fails in a physical bus network?
• **1.6.2** What logical topology does a physical bus network use?
• **1.6.3** Which physical topology should be used when rewiring the office?

Q1.7 [6 Marks]: Explain the difference between error detection (Parity bits) and error correction (Hamming Code).

PART B: TERM TEST 2 & SPECIAL/SICK TEST 2 PAST QUESTIONS (2024 & 2026)

1. SCOB032 Main Test 2 (2026) [50 Marks]

Q1 [15 Marks — Enterprise VLSM Design]: Provided address space **10.10.0.0/16** to design a scheme for LAN1 (300 hosts), LAN3 (100 hosts), LAN4 (55 hosts), LAN2 (30 hosts), and WAN1 (2 hosts).
• **1.1 [2 Marks]:** State the number of host bits available in 10.10.0.0/16 with calculations.
• **1.2 [3 Marks]:** Determine the smallest CIDR prefix for each requirement.
• **1.3 [6 Marks]:** Allocate address blocks in **descending order** using VLSM. List Network Address, First Usable IP, Last Usable IP, and Broadcast Address for each subnet.
• **1.4 [2 Marks]:** Determine the first unallocated IP address remaining after completing allocations.
• **1.5 [2 Marks — The Technician Trap]:** A technician assigns 10.10.2.150 to a device in LAN3. Critically evaluate whether this address belongs to LAN3 using your calculated ranges.

Q2 [11 Marks — Transport & Application Layers]:
• **2.1 [5 Marks]:** Host A (ISN_A = 1,543) establishes a TCP connection with Host B (ISN_B = 2,780). (i) Draw a labeled 3-way handshake diagram. (ii) Determine the ACK Number in Host B's SYN-ACK response. (iii) Determine the Sequence and ACK Numbers in Host A's final ACK segment.
• **2.2 [6 Marks]:** Match Service A (web browser viewing course info), Service B (uploading research datasets to a file server), and Service C

(sending official email) to application protocols and standard ports from the list: 21, 22, 25, 53, 80.

Q3 [24 Marks — Mobile Handoff & Security]:
• **3.1 [7 Marks]:** A student downloading course materials moves between cellular base stations. (i) Identify the process allowing active connection transfer between base stations. (ii) Explain why this process is necessary. (iii) State two factors affecting mobile connection quality during transit.
• **3.2 [6 Marks]:** Diagram showing Asymmetric Encryption across nodes A, M, and B. Identify components A, M, and B, and explain how message secrecy is maintained.
• **3.3 [11 Marks]:** Registrar database flooded during registration. (i) Identify the network attack and affected CIA Triad principle (Availability). (ii) Explain how this attack causes service unavailability. (iii) Recommend two security controls to mitigate this attack.

2. SCOB032 Special / Sick Test 2 (August 2026) [50 Marks]

Q1 [10 Marks — VLSM Allocation]: Allocated **172.20.0.0/23** for Administration (120 hosts), Finance (60 hosts), Research (28 hosts), and Security (12 hosts).
• **a)** Determine the smallest suitable CIDR prefix for Administration.
• **b)** Allocate subnets in descending order using VLSM; list Network Address and Broadcast Address for each.
• **c)** Determine the first unallocated IP address remaining.

Q2 [10 Marks — Handshake Math & Port Services]:
• ISN_Student = 4,250; ISN_Server = 7,600. Determine Sequence and ACK numbers for SYN-ACK and final ACK.
• Identify application protocols and ports for: Accessing a secure website (HTTPS/443), Domain name resolution (DNS/53), Secure remote Linux login (SSH/22).

Q5 [10 Marks — Network Troubleshooting]:
• Problem A: Workstation has valid local link but cannot reach remote server.
• Problem B: Damaged cable prevents signals from reaching workstation.
• Problem C: Switch receives frame whose destination MAC is not in its table.
• Identify relevant OSI layers, networking devices responsible, and explain switch flooding behavior for Problem C.

3. SCOB032 Term Test 2 (2024) [50 Marks]

Q1.1 [11 Marks]: Explain CSMA/CA vs CSMA/CD. Diagram the Hidden Terminal Problem and RTS/CTS mitigation.

Q1.2 [5 Marks]: Identify protocol suitable for satellite networks with long propagation delays.

Q1.3 [8 Marks — Class B Subnetting]: Subnet 172.16.0.0 into 8 subnets.
• **i)** Determine subnet mask and usable host capacity.
• **ii)** Calculate Subnet Address, Broadcast Address, First Usable IP, and Last Usable IP for Subnet 3.
• **iii)** Explain how subnetting prevents IP conflicts.

Q1.4 [16 Marks]: High BER in 200 km fiber link using WDM/QAM (causes and solutions); Passive vs Active attacks; Direct vs Indirect routing diagrams.

PART C: FINAL MAIN & SUPPLEMENTARY EXAMINATION PAST QUESTIONS (2016–2024)

1. SCOB032 Main Exam (OCT/NOV 2024) [60 Marks]

Q1.1 [4 Marks]: Describe first four OSI layers and encapsulation.

Q1.2 [6 Marks]: Sekhukhune municipality coaxial/twisted-pair network failing during thunderstorms (causes and cable-based fiber solution).

Q1.3 [6 Marks]: Compare Simplex, Half-Duplex, and Full-Duplex modes.

Q1.5 [7 Marks]: Mobile workforce direct vs indirect routing diagrams, home/foreign agent roles, and latency comparison.

Q1.6 [4 Marks]: MAN spanning 8 buildings using 164.10.0.0 (255 hosts per building). Calculate subnet mask, network address, broadcast address, and host range for the second building.

Q2.1 [6 Marks]: CSMA/CD operation on wired Ethernet and impact of traffic increases on TCP/IP applications.

Q2.2 [4 Marks]: Diagram the Hidden Terminal Problem.

Q2.3 [5 Marks]: Evaluate ALOHA vs Polling protocols for a new office LAN.

Q2.5 [3 Marks]: Dynamic Source Routing (DSR) in a MANET.

2. SCOB032 Main Exam (OCT/NOV 2023) [60 Marks]

Q1.1 [6 Marks]: Communication mode selection for public address system vs. graphic designers sharing files.

Q1.2 [6 Marks]: Explain OSI layers excluding Application.

Q1.6 [3 Marks]: Define subnetting, purpose, and benefits.

Q1.7 [2 Marks]: Convert 222.1.1.20 with mask 255.255.255.192 and 135.1.1.25 with mask 255.255.248.0 to CIDR.

Q1.8 [3 Marks]: Complete conversion table for InterNIC prefixes /12, /20, /30 to Subnet Mask and Usable Hosts.

Q2.1 [10 Marks]: Subnet 192.168.0.0/24 for HR (30), Sales (25), IT (15), and Future Growth (50). Design topology and allocate subnets.

Q2.2 [3 Marks]: TCP 3-way handshake diagram.

Q2.5 [4 Marks]: Asymmetric Public Key Cryptography encryption/decryption diagram.

Q2.6–2.8 [6 Marks]: Zero-Day vulnerabilities, cyberattacks, and proactive security controls.

3. Historical Exam Highlights (2016 & 2019)

2016 Q1.1 [8 Marks]: Define throughput and state 6 factors determining throughput.

2016 Q1.3 [7 Marks]: CSMA/CD operation and TCP/IP layer mapping.

2019 Q3.1 [4 Marks]: Derive class ranges for binary IP addresses by inspecting leading high-order bits.

2019 Supp Q3.1 [6 Marks]: Given 198.168.4.121/27, calculate Network ID, Broadcast ID, and total usable hosts.